Capacity-feasible CVRP clustering on an analog neutral-atom register (Track A)

Pasqal challenge — team DuQuantum — repository 'qhack'

2026-10-25

1. Model

What we chose, why, and what it costs. Cluster-first / route-second: the register decides *which* customers share a vehicle; each cluster is then routed exactly (Held–Karp). The full (customer, vehicle, position) encoding needs $\sim N^2K$ variables — 32 / 300 / 900 on S / M / L — against 80 atoms (DERIVATIONS Q1); clustering needs NK. The price is a *decomposition gap* (our objective's best clustering is not always the CVRP optimum), reported in §5.

Variables and objective (brief §2). Customers $i = 1..N$ with demands d_i, K identical vehicles of capacity Q, Euclidean costs d_{ij}. One-hot $x_{ik} \in \{0,1\}$ (customer i on vehicle k), $n = NK$ atoms: **8 / 30 / 60** on S / M / L, no slack. Objective: total pairwise distance inside each cluster, $f = \sum_k \sum_{i<j} d_{ij} x_{ik} x_{jk}$, additive over clusters, so an exact set-partition DP (`cvrp/exact.py`) certifies the *model* optimum on every tier.

Constraints → penalties (DERIVATIONS §4, `docs/MODEL.md`). Exactly-one [=]: squared penalty $A \sum_i (1 - \sum_k x_{ik})^2 \to h = -A$ on every variable, $J = 2A$ on each customer's K-clique. Capacity [≤]: unbalanced penalization $\sum_k [-\lambda_1 (Q - L_k) + \lambda_2 (Q - L_k)^2]$ (Montañez-Barrera et al. 2024), expanded. Its per-customer linear part $c_i \sum_k x_{ik}$ is a *constant* on one-hot strings (the one-hot identity), so it is dropped: λ_1 disappears and what remains is $+2\lambda_2 d_i d_j$ on same-vehicle pairs, a **soft load-balance term** $\lambda_2 \sum_k L_k^2$. For $K = 2$ its optimum is capacity-feasible whenever a feasible split exists; for $K \geq 3$ only likely — so we score feasible decodes only and report the feasibility rate (Pitfall 9.1).

What the device sees. At $\Omega = 0$ the final Hamiltonian is a QUBO with one *uniform* linear coefficient (δ_f) and *positive* couplings (C_6/r^6) — exactly the pattern above: every $J > 0$, every $h = -A$. Weights: $A = \max_i \sum_{j \neq i} (d_{ij} + 2\lambda_2 d_i d_j) + 1$ (provable: every argmin is one-hot), $\lambda_2 = 0.15\, R/(2Q)$ from a 40-seed brute-force scan. Checker on tier S (`out/gate1/`): argmin feasible and optimal on 5/5 shipped instances; A survives one halving and fails at ¼; λ_2 is not binding on these instances.

2. Variable accounting (brief Derive 4.1)

Encoding	count	S	M	L	fits 80?
assignment variables only	NK	8	30	60	yes
+ binary slack (textbook)	$NK + K\lceil \log_2(Q{+}1) \rceil$	14	45	80	yes (L = cap)
ours (`onehot-ub`)	NK	**8**	**30**	**60**	yes

(`scripts/variable_count.py`; textbook rows implemented and checked, DERIVATIONS §3.) Saving: 6 / 15 / 20 atoms; no discarded readouts; the capacity term adds no coupling pairs beyond the objective's.

3. Embedding (DERIVATIONS §5)

What we chose, why, and what it costs. A generic least-squares placement cannot realise the one-hot rook's graph, so we fit the register on the one-hot subspace only (§5.7). For $K = 2$ (tier S) the fit realises the model; for $K = 3$ (tier M) exactly-one assignments are produced with ≈ 98–100 % probability while capacity feasibility (4–45 %) and objective quality remain unresolved (§5, DERIVATIONS §6.7, Q-M3).

Why the generic embedder fails. The K copies of a customer form a $2A$-clique 5 µm apart, so a neighbouring customer's atom is at nearly the same distance from all K copies: the "zero" cross-vehicle couplings come out as large as the wanted ones. Measured: relative coupling error 0.95 ($\|U/\alpha - J\|_F / \|J\|_F$ over all $i \neq j$, lower is better; `cvrp/embed.py` `FidelityReport`), always reported with the worst-pair error $\max_{J_{ij}>0} |U_{ij}/\alpha - J_{ij}|/J_{ij}$ — zero-target pairs are excluded from it and enter separately as the largest spurious coupling $\max_{J_{ij}=0} U_{ij}/\alpha$ — and, on S_seed04, an infeasible device ground state (model optimum at rank 9).

Ask for less. On one-hot strings each customer pair contributes exactly one coupling, so only *same-vehicle minus cross-vehicle* must equal αw_{ij} (identity 5.7b); cross couplings need only be constant. Each customer becomes a rigid 5 µm K-gon (clique at the minimum separation, $\alpha = C_6/5^6/2A$); centres and orientations are fitted (L-BFGS, 6 seeded starts) so the device energies of all one-hot strings match αE_{QUBO}, with explicit single-flip defect gaps and a fitted δ_f (a constant on the subspace: it only sets the defect gaps). Registers are cached and committed (`embeddings/`). Per run:

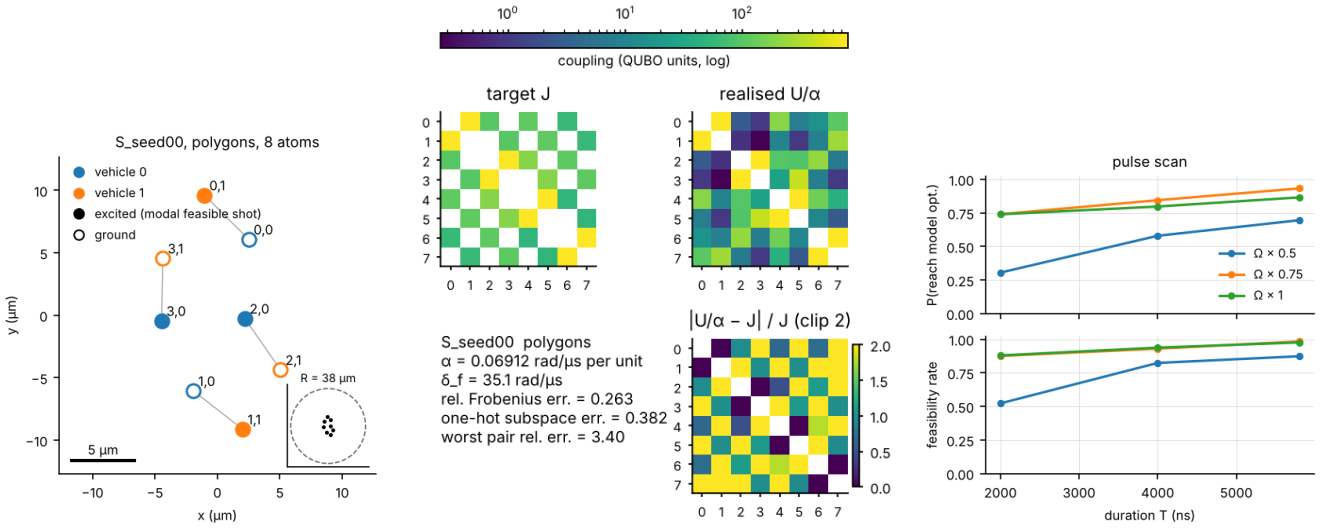


Figure 1: Left: S_seed00 register; each 5 µm dimer is one customer's exactly-one clique, filled atoms = excited atoms of the modal feasible shot (one per dimer: a one-hot string). Middle: target vs realised couplings; clique entries exact, cross-dimer entries not — the full-matrix coupling error (0.263) is dominated by what the fit does not need (worst pair 3.40), the subspace error (0.382) by what it does. Right: pulse scan, three tier-S instances (in-house SV, 500 shots, mean): both curves rise with duration at every Ω; 0.75 $\Omega_{\max}$ is on top at 4000 and 5800 ns; Ω/2 loses feasibility.

full-matrix and subspace coupling error, and the *realised-register check* (device argmin feasible and optimal? smallest defect gap). Tier S: 5/5 yes.

Where it breaks (tier M, $K = 3$). Triangles 5 µm on a side cannot be far from each other's neighbours: the subspace-fit residual is 38 (S: 0.06–0.32) and the model optimum ranks ~10 000th of 59 049 one-hot strings; opening the triangles to 6–10 µm does not fix the ordering (rank ≥ 1 279; DERIVATIONS §5.7) — the brief's §4 warning in its sharpest form, a property of one-hot encodings on a 2-D $1/r^6$ register. The cliques themselves do embed, under a bound found on M (§6.7): at $\Omega = 0$ a K-gon's one-hot state is its ground state only if $0 < \delta_f < U_{\min}$, the coupling of the *weakest* intra-polygon pair. For $K \leq 3$ every pair is the 5 µm side, $U_{\min} = U_{\text{clique}} = 55.1$ rad/µs; S fits land at 0.49–0.68 U, the ill-conditioned M fit ran to the device maximum 125.66 (2.28 U), so the embedder now enforces the bound (fallback 0.65 $U_{\min}$). For $K = 4$ (tier L) a 5 µm square has 7.07 µm diagonals with $U_{\text{side}}/8 = 6.9$ rad/µs, so the L registers run at $\delta_f = 4.5$ rad/µs (results.csv: 4.475); the first L run (δ_f 38.7) was killed before reporting (DID_NOT_WORK). Four points in a plane are never equidistant (max/min distance ≥ √2, a factor 8 in $1/r^6$), so a planar one-hot clique for $K \geq 4$ is never uniform.

4. Pulse (DERIVATIONS §6)

Chosen: $T = 5800$ ns (device maximum minus 200 ns), $\Omega = 0.75\,\Omega_{\max}$, fitted δ_f (35–38 rad/µs on S, inside ±125.66). Shape: three-point Ω ramp (15 % / 70 % / 15 % of T), linear δ sweep from $\delta_0 = -\min(\max(|\delta_f|, 2\Omega_{\max}), \delta_{\max})$ to δ_f while Ω is on its plateau; every sequence is built with Pulser on `AnalogDevice` and validated (`to_abstract_repr()`, limits from the device object). Why: a scan on three tier-S instances (Fig. 1 right, §6.6), scored by P(reach), best gap and feasibility, never mean cost: longer is monotonically better (2000 → 4000 → 5800 ns at every Ω); $0.75\,\Omega_{\max}$ beats full and half power at 4000 and 5800 ns; $\delta_f \times \{0.8, 1, 1.25\}$ peaks at the fitted value. S_seed02/03, not used in the scan, reach the model optimum in 99 % / 96 % of shots with these settings. Emulators: pulser-simulation (qutip) for every S headline row; emu-sv cross-check (S_seed00: 0.96 vs 0.94); an in-house exact state-vector integrator from Pulser's own samples for the scans (TV 0.05–0.07 from qutip at 500 shots = sampling noise); emu-mps for M/L, bond dimension recorded.

5. Results (brief Table 3; `results.csv`, `python -m cvrp.results`)

Reading the table. One row per (instance, method); raw shots, no repair. *feas.*: fraction of shots decoding to a capacity-feasible assignment. *model gap*: best feasible shot vs the exact clustering-model optimum (set-partition DP); *CVRP gap*: that shot routed exactly vs C^* (proven on S, best known on M/L) — both on feasible decodes only, "undefined" if none. *P(reach C^*)*: shots whose clustering routes to C^* (includes the decomposition loss); *P(gap ≤ 5 %)*: shots within 5 % of the model optimum — equal to P(reach model optimum) on every S instance and on M_seed00/01 (smallest non-zero feasible gap 11–62 %), not on M_seed02 (6 feasible assignments within 5 %, smallest 0.98 %; Q-M3). *fidelity*: relative coupling error $\|U/\alpha - J\|_F / \|J\|_F$ over the full matrix, lower is better (§3). Rows: **ours** = headline (`onehot-ub`, polygon register, 5800 ns / 0.75 $\Omega_{\max}$; qutip on S, emu-mps on M); **tight A** = ablation; **random assignments (one-hot)** = fair control (assignment-valid random strings, same decoder); **random sampler** = uniform bitstrings; **qubo-solver quantum / tabu** = Pasqal's library on our QUBO, its own embedding and drive.

instance	method	feas.	model gap	CVRP gap	P(reach C*)	P(gap≤5%)	fidelity	backend
S_seed00	ours	1.00	0.0 %	0.0 %	0.94	0.94	0.26	qutip
S_seed00	ours, tight A (ablation)	1.00	0.0 %	0.0 %	0.03	0.03	0.31	qutip
S_seed00	random assignments (one-hot)	0.63	0.0 %	0.0 %	0.12	0.12	0.26	random-onehot
S_seed00	qubo-solver quantum	1.00	0.0 %	0.0 %	0.78	0.78	0.12	qs-quantum
S_seed00	qubo-solver tabu	1.00	0.0 %	0.0 %	1.00	1.00	undef.	qs-classical
S_seed00	random sampler	0.04	0.0 %	0.0 %	0.01	0.01	0.95	random
S_seed01	ours	0.95	0.0 %	0.0 %	0.95	0.95	0.21	qutip
S_seed01	ours, tight A (ablation)	0.52	0.0 %	0.0 %	0.52	0.52	0.29	qutip
S_seed01	random assignments (one-hot)	0.25	0.0 %	0.0 %	0.12	0.12	0.21	random-onehot
S_seed01	qubo-solver quantum	0.80	0.0 %	0.0 %	0.78	0.78	0.16	qs-quantum
S_seed01	qubo-solver tabu	1.00	0.0 %	0.0 %	1.00	1.00	undef.	qs-classical
S_seed01	random sampler	0.02	0.0 %	0.0 %	0.01	0.01	0.94	random
S_seed02	ours	1.00	0.0 %	0.0 %	0.99	0.99	0.31	qutip
S_seed02	qubo-solver quantum	0.95	0.0 %	0.0 %	0.18	0.18	0.21	qs-quantum
S_seed02	qubo-solver tabu	1.00	0.0 %	0.0 %	1.00	1.00	undef.	qs-classical
S_seed02	random sampler	0.04	0.0 %	0.0 %	0.01	0.01	0.31	random
S_seed03	ours	1.00	0.0 %	15.6 %	0.01	0.96	0.29	qutip
S_seed03	qubo-solver quantum	0.99	0.0 %	15.6 %	0.18	0.79	0.14	qs-quantum
S_seed03	qubo-solver tabu	1.00	0.0 %	15.6 %	0.00	1.00	undef.	qs-classical
S_seed03	random sampler	0.05	0.0 %	15.6 %	0.01	0.01	0.29	random
S_seed04	ours	1.00	0.0 %	21.7 %	0.02	0.89	0.32	qutip
S_seed04	ours, tight A (ablation)	0.99	0.0 %	21.7 %	0.02	0.94	0.34	qutip
S_seed04	random assignments (one-hot)	0.62	0.0 %	21.7 %	0.11	0.13	0.32	random-onehot
S_seed04	qubo-solver quantum	1.00	27.3 %	30.0 %	0.00	0.00	0.19	qs-quantum
S_seed04	qubo-solver tabu	1.00	0.0 %	21.7 %	0.00	1.00	undef.	qs-classical
S_seed04	random sampler	0.04	0.0 %	21.7 %	0.01	0.01	0.95	random
M_seed00	ours (emu-mps)	0.04	67.9 %	29.9 %	0.00	0.00	0.58	emu-mps χ=32
M_seed00	qubo-solver quantum	0.02	72.1 %	35.2 %	0.00	0.00	0.09	qs-quantum χ=32
M_seed00	random sampler	0.00	undef.	undef.	0.00	0.00	0.58	random
M_seed01	ours (emu-mps)	0.39	37.9 %	9.2 %	0.00	0.00	0.08	emu-mps χ=32
M_seed01	qubo-solver quantum	0.37	47.5 %	11.2 %	0.00	0.00	0.08	qs-quantum χ=32
M_seed01	qubo-solver tabu	1.00	65.4 %	17.6 %	0.00	0.00	undef.	qs-classical
M_seed01	random sampler	0.00	undef.	undef.	0.00	0.00	0.08	random
M_seed02	ours (emu-mps)	0.45	23.0 %	16.0 %	0.00	0.00	0.47	emu-mps χ=32
M_seed02	qubo-solver quantum	0.39	34.2 %	13.6 %	0.00	0.00	0.09	qs-quantum χ=32
M_seed02	qubo-solver tabu	1.00	10.2 %	2.8 %	0.00	0.00	undef.	qs-classical
M_seed02	random sampler	0.00	undef.	undef.	0.00	0.00	0.47	random
L_seed00	random sampler	0.00	undef.	undef.	0.00	0.00	0.65	random
L_seed01	random sampler	0.00	undef.	undef.	0.00	0.00	0.65	random
L_seed02	random sampler	0.00	undef.	undef.	0.00	0.00	0.64	random

(generated by `python scripts/report_tables.py` from `results.csv`; raw shots only — repair rows are in results.csv under their own label; shots: 500 per emulator row, 2000 for the random sampler, 1 for a classical solver)

Reference C^* (brief §9.3). CPLEX is not importable, so C^* is the best of (i) qubo-solver's SA (stated parameters) on our penalized `onehot-ub` QUBO and (ii) our constrained SA on the CVRP, each 10 seeded restarts, **60 s max solve time** (`instances/*.json`). On tier S the exhaustive set-partition DP is also run and agrees 5/5, which is the brief's level 1, so S references are `proven_optimum`; M/L are `best_known`; the CPLEX cross-check stays an outstanding item (DID_NOT_WORK). The QUBO-side SA reaches C^* on 5/5 S and 2/3 M instances but stays 23–34 % above it on all three L (599.9 / 554.6 / 521.7 vs 447.0 / 434.9 / 424.7; Q-ref): the penalized one-hot QUBO is harder to search than the constrained problem, so C^* is never the QUBO solver alone. Decomposition gap (model optimum routed exactly vs C^*):

instance	ρ	C* (best known)	model optimum routed exactly	decomposition gap
S_seed00	0.714	262.97	262.97	0.0 %
S_seed01	0.857	245.02	245.02	0.0 %

instance	ρ	C* (best known)	model optimum routed exactly	decomposition gap
S_seed02	0.714	254.62	254.63	0.0 %
S_seed03	0.643	197.44	228.34	15.6 %
S_seed04	0.714	184.42	224.49	21.7 %
M_seed00	0.950	432.00	432.00	0.0 %
M_seed01	0.733	368.36	368.36	0.0 %
M_seed02	0.700	360.86	385.62	6.9 %
L_seed00	0.650	446.97	525.71	17.6 %
L_seed01	0.708	434.91	470.67	8.2 %
L_seed02	0.600	424.66	480.17	13.1 %

Tier S, in four steps. (1) *Register.* On every instance the realised register's own argmin is the model optimum (device rank 1); relative coupling error 0.21–0.32 vs 0.94–0.95 for the generic embedder, which put the optimum at rank 9 on S_seed04. (2) *Feasibility.* 0.95–1.00 of raw shots are capacity-feasible (generic register: 0.06–0.22; uniform random: 0.02–0.05). (3) *P(reach) vs controls — useful quantum work (brief §9.2).* The model-optimal clustering is returned in 89–99 % of raw shots on all five instances. The fair control, `random-onehot`, reaches it with P = 2/16 ≈ 0.12 on every instance (two labelings of one optimal clustering among 16 assignments; feasible 25–63 %); the headline rows give 0.94 / 0.95 / 0.89 on S_seed00/01/04 at equal decoding — a 7–8× factor, so the register, not the decoder, does the work (uniform bitstrings: 0.7–1.2 %; generic embedder: 3–13 %). qubo-solver's own pipeline (BLaDE + its drive shaper, 5940 ns): 0.18–0.79 on four instances, 27 % model gap on S_seed04; its tabu search finds every model optimum (its default SA, a single-start greedy descent at $T_0 = 5$, does not; we report a stated-parameter SA instead). (4) *Decomposition.* Where P(reach C^*) is low with a 0 % model gap (S_seed03/04: 0.01 / 0.02) the loss is the decomposition gap (15.6 % / 21.7 %; M_seed02: 6.9 %): the closeness objective's optimum is not the CVRP optimum — the cost of cluster-first (§1), not of the device.

Ablation — the exactly-one margin is physically load-bearing. `onehot-ub-tight` replaces $A = \max_i \sum_j w_{ij} + 1$ by the tight bound $A = R_w/K\,(1 + 1\%)$ (still sound, checker-verified), same register construction, pulse tuned for the original A: P(reach model optimum) 0.94 → 0.03 on S_seed00 (device ground state = third-best clustering), 0.95 → 0.52 on S_seed01, 0.89 → 0.94 on S_seed04 (qutip; in-house SV 0.03 / 0.56 / 0.96): a smaller A shrinks the single-flip defect gaps the sweep must resolve and leaves less margin for the embedding error. The headline stays `onehot-ub`.

Tier M — exactly-one assignments are produced with ≈ 98–100 % probability; capacity feasibility (4–45 %) and objective quality remain unresolved (DERIVATIONS §6.7, Q-M2, Q-M3). *Before:* our first M rows had $\delta_f = 125.66$ rad/µs $= 2.28\,U_{\text{clique}}$, so two excitations per triangle were favoured and 0 of 500 shots were decodable (kept as `|pre-6.7` in results.csv) — a wrong sweep end point, not a feasibility limit of the encoding. *After* (same registers, bounded δ_f, emu-mps $\chi = 32$, 500 shots; rows "ours (emu-mps)"): decodable 0 → 0.994 / 0.984 / 0.996, feasible 0 → 0.044 / 0.388 / 0.446 on M_seed00/01/02, best model gap 67.9 / 37.9 / 23.0 %, CVRP gap 29.9 / 9.2 / 16.0 %. A uniform random sampler decodes 0 of 2000 shots on every M instance (table; M_seed01 re-measured on the cached register), so the one-hot shots are the device's own work. *Where the remaining loss sits* (Q-M3, all 59 049 one-hot strings enumerated, `docs/checks/landscapes.md`): (i) *penalty* — the load-balance term leaves the model optimum at QUBO rank 1 on every S instance and on M_seed00/01, but on M_seed02 the QUBO argmin is a feasible assignment 10.2 % above the model optimum (rank 331) — exactly qubo-solver's tabu row — so ~10 of M_seed02's 23 points of model gap are the λ_2 rule, not the register; (ii) *register* — the optimum's device rank among one-hot strings is 10 474 / 12 966 / 4 192 and is the main loss; (iii) *sweep* — the emulated best feasible shots (67.9 / 37.9 / 23.0 %) beat the register's own best feasible one-hot string under E_{dev} (82.0 / 101.3 / 93.9 %), so the pulse is not the bottleneck. qubo-solver's independent pipeline (BLaDE register, $\chi = 32$) lands in the same regime on all three M (feasible 0.016 / 0.37 / 0.388, decodable 0.99 / 0.998 / 0.992, model gap 72.1 / 47.5 / 34.2 %, CVRP gap 35.2 / 11.2 / 13.6 %), while its tabu search on the same QUBO reaches 17.6 % / 2.8 % above C^* on M_seed01/02: the QUBO is solvable classically; the planar register limits it. A $K = 2$ bisection does not help (GATE2_OPTIONS (f): rank 192 594 / 12 217).

Tier L — measured rows pending. Three 60-atom registers were built with the $K = 4$ bound ($\delta_f = 4.475$ rad/µs); the model optimum sits at one-hot rank 16 534 / 1 063 / 14 862 of 1 048 576 (results.csv, random rows), and the uniform random control decodes 0 of 2000 shots on all three; the decomposition gap is already known (17.6 / 8.2 / 13.1 %, table above). Emulations (emu-mps $\chi = 32$) and an M_seed00 $\chi = 64$ convergence run are running; rows are reported as measured.

6. What did not work

`docs/DID_NOT_WORK.md`, one sentence each: no PyPI from the sandboxes; CPLEX not importable; a typed number caught by review; qubo-solver's default SA; a blocking calibration job; a weight scan with no trend (the embedder's local minimum moves P(reach) more than the weights); a non-reproducible embedder (fixed by committing registers); the tight-A ablation; an unbounded δ_f that made the first tier-M rows undecodable (§6.7); the triangle bound applied to tier-L squares (first L run killed before reporting); a `|` in a run tag that lost one Windows record; the polygon optimiser giving two registers from one seed when runs overlap (only cache-loaded registers are reported now); the CPLEX cross-check, still outstanding; and the one-hot *ordering* not embedding for $K = 3$ — the limit of this encoding on this device.

7. Use of AI

`docs/AI_USE.md`. Claude wrote the code, plumbing and most prose from the team's decisions; the Gate-1 formulation (`docs/MODEL.md`) was AI-drafted at the team's request under time pressure, checked by brute force, and must be re-derived by the team before judging. Every number here is produced by a script from committed records.